

Practice Questions →

$$(I) x^2 - 43x + 450 = 0$$

$$\text{Product} = 450$$

$$\text{Sum} = 43$$

→ Both roots positive

$$(+18, +25) \times 5$$

$$(90, 125)$$

$$(II) 5y^2 - 92y + 36 = 0$$

$$\text{Product} = 36 \times 5 = 90$$

$$\text{Sum} = 92$$

→ Both Roots positive

$$(\overline{90}, \overline{2})$$

Comparison of Roots →

$$\left. \begin{array}{l} \text{Roots of } x = 90, 125 \\ \text{Roots of } y = 90, 2 \end{array} \right\} \begin{array}{ll} 90 = 90 & 125 > 90 \\ 90 > 2 & 125 > 2 \end{array}$$

$$\boxed{\text{So, } x \geq y}$$

$$(I) 2x^2 - 29x + 60 = 0$$

$$\text{Sum} = 29$$

$$\text{Product} = 180$$

→ Both roots positive

$$\left(\frac{24}{2}, \frac{5}{2} \right)$$

$$(12, 2.5)$$

$$(II) y^2 + 3y - 4 = 0$$

$$\text{Product} = 4$$

$$\text{Difference} = 3$$

→ opposite roots
smaller root positive

$$(-4, +1)$$

Comparison of Roots →

$$\left. \begin{array}{l} \text{Roots of } x = 12, 2.5 \\ \text{Roots of } y = -4, +1 \end{array} \right\} \begin{array}{ll} 12 > -4 & 2.5 > -4 \\ 12 > 1 & 2.5 > 1 \end{array}$$

$$\boxed{\text{So, } x > y}$$

$$(I) x^2 - 32x + 252 = 0$$

Product = 252

Sum = 32

→ Both roots positive

(18, 14)

$$(II) y^2 + 19y - 252 = 0$$

Product = 252

Difference = 19

→ Smaller root positive

(-28, +9)

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = 18, 14 \\ \text{Roots of } y = -28, 9 \end{array} \left\{ \begin{array}{ll} 18 > -28 & 14 > -28 \\ 18 > 9 & 14 > 9 \end{array} \right\}$$

So, $x > y$

$$(I) 3x^2 - 13x + 14 = 0$$

(+, +)

Both roots positive

So, $x > y$

$$(II) 2y^2 + 7y + 3 = 0$$

(-, -)

Both roots negative

{ Sign Method }

$$(I) x^2 + 10x - 8075 = 0$$

Product = 8075

Difference = 10

(-95, 85)

$$(II) y^2 - 178y + 7920 = 0$$

Product = 7920

Sum = 178

(90, 88)

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = -95, 85 \\ \text{Roots of } y = 90, 88 \end{array} \left\{ \begin{array}{ll} -90 < 90 & 85 < 90 \\ -90 < 88 & 85 < 88 \end{array} \right\}$$

So, $x < y$

$$(I) \quad 2x^2 + 13x + 15 = 0$$

(-, -)

Both Roots negative

$$\boxed{\text{So, } x < y}$$

$$(II) \quad 5y^2 - 11y + 12 = 0$$

(+, +)

Both roots positive

{ Due to
Sign Method }

$$(I) \quad 2x^2 - 15x + 28 = 0$$

(+, +)

Both roots positive

$$\boxed{\text{So, } x > y}$$

$$(II) \quad 3y^2 + 16y + 5 = 0$$

(-, -)

Both roots negative

{ Sign Method }

$$(I) \quad 24x^2 - 155x + 19 = 0$$

$$\text{Product} = (24)(19)$$

$$= 456$$

$$\text{Sum} = 155$$

Both roots positive

$$\left(\frac{152}{24}, \frac{3}{24} \right) \times 120$$

← coefficient
of y^2

$$(152 \times 5, 3 \times 5)$$

$$(760, 15)$$

$$(II) \quad 120y^2 - 22y + 1 = 0$$

$$\text{Product} = 120$$

$$\text{Sum} = 22$$

Both roots positive

$$\left(\frac{12}{120}, \frac{10}{120} \right)$$

Comparison :

$$\text{Roots of } x = 760, 15$$

$$\text{Roots of } y = 12, 10$$

$$\boxed{\text{So, } x > y}$$

$$(I) 3x^2 - 47x + 30 = 0$$

$$\text{Product} = 90$$

$$\text{Sum} = 47$$

$$(+45, +2)$$

$$(II) y^2 - 31y + 240 = 0$$

$$\text{Product} = 240$$

$$\text{Sum} = 31$$

$$(+15, +16) \times 3$$

$$(45, 48)$$

Comparison of Roots $\rightarrow$

$$\text{Roots of } x = 45, 2$$

$$\text{Roots of } y = 45, 48$$

$$\left\{ \begin{array}{ll} 45 = 45 & 2 < 45 \\ 45 < 48 & 2 < 48 \end{array} \right\}$$

$$\boxed{\text{So, } x \leq y}$$

$$(I) 3x^2 - 13x + 14 = 0$$

$$(II) 2y^2 + 7y + 3 = 0$$

Roots (+, +)

Roots (-, -)

$$\boxed{x > y}$$

Sign Method

$$(I) 5x^2 - 3x - 2 = 0$$

$$\text{Product} = 10$$

$$\text{Difference} = 3$$

$$(5, -2) \times 9$$

$$(45, -18) \text{ Roots}$$

$$(2) 9y^2 - 19y + 10 = 0$$

$$\text{Product} = 90$$

$$\text{Sum} = 19$$

$$(10, 9) \times 5$$

$$(50, 45) \text{ Roots}$$

Comparison of Roots $\rightarrow$

$$\text{Roots of } x = 45, -18$$

$$\text{Roots of } y = 50, 45$$

$$\left\{ \begin{array}{ll} 45 < 50 & -18 < 50 \\ 45 = 45 & -18 < 45 \end{array} \right\}$$

$$\boxed{\text{So, } x \leq y}$$

$$(I) 4x + 3y = -29$$

$$(II) -8x + 9y = -47$$

Elimination Method $\rightarrow$

$$(4x + 3y = -29) \times 2$$

$$-8x + 9y = -47$$

— Eq 1

— Eq 2

$$\begin{array}{r} 8x + 6y = -58 \\ -8x + 9y = -47 \\ \hline \end{array}$$

(Addition)

$$15y = -105$$

$$y = -7$$

Put value of y in Eq(1)

$$4x + 3(-7) = -29$$

$$4x = -29 + 21$$

$$4x = -8$$

$$x = -2$$

$$\boxed{\begin{array}{l} x = -2 \\ y = -7 \end{array}}$$

$$(I) 3x^2 - 14x - 5 = 0$$

Product = 15

Difference = 14

$$(15, -1) \times \cancel{x}$$

$$(II) 3y^2 + 7y + 2 = 0$$

Product = 6

Sum = 7

$$(-6, -1) \times \cancel{x}$$

Comparison of Roots $\rightarrow$

$$\text{Roots of } x = 15, -1$$

$$\text{Roots of } y = -6, -1$$

$$\left. \begin{array}{l} 15 > -6 \\ -1 > -6 \\ 15 > -1 \\ -1 = -1 \end{array} \right\}$$

$$\boxed{\text{So, } x \geq y}$$

$$(I) 15x^2 - 53x + 42 = 0$$

$$\text{Product} = 42 \times 15 = 630$$

$$\text{Sum} = 53$$

→ Both Roots positive

$$(18, 35) \times 20$$

$$(360, 700)$$

$$(II) 20y^2 - 49y + 30 = 0$$

$$\text{Product} = 600$$

$$\text{Sum} = 49$$

→ Both roots positive

$$(25, 24) \times 15$$

$$(375, 360)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = 360, 700 \\ \text{Roots of } y = 375, 360 \end{array} \left\{ \begin{array}{ll} 360 < 375 & 700 > 375 \\ 360 = 360 & 700 > 360 \end{array} \right\}$$

Hence, No Relation can be established.

Roots :-

$$(I) x^2 - 7\sqrt{5}x + 50 = 0$$

We will modify the eqn.

$$x^2 - 7x + \frac{50}{5} = 0$$

$$x^2 - 7x + 10 = 0$$

$$\text{Product} = 10$$

$$\text{Sum} = 7$$

$$(5, 2) \Rightarrow (5\sqrt{5}, 2\sqrt{5})$$

$$(II) y^2 + y - 20 = 0$$

$$\text{Product} = 20$$

$$\text{Difference} = 1$$

$$(+4, -5)$$

Comparison →

Squaring all roots

$$\Rightarrow \begin{array}{l} x = 125, 20 \\ y = 16, -25 \end{array} \left\{ \begin{array}{ll} 125 > 16 & 20 > 16 \\ 125 > -25 & 20 > -25 \end{array} \right\}$$

$$\boxed{x > y}$$

$$(I) x^2 - 8\sqrt{2}x + 30 = 0$$

Modify the eqⁿ.

$$x^2 - 8x + \frac{30}{2} = 0$$

$$x^2 - 8x + 15 = 0$$

$$\text{Product} = 15$$

$$\text{Sum} = 8$$

$$(+3, +5) \Rightarrow 3\sqrt{2}, 5\sqrt{2}$$

$$(II) y^2 - 2\sqrt{2}y - 6 = 0$$

Modify the equation

$$y^2 - 2y - \frac{6}{2} = 0$$

$$y^2 - 2y - 3 = 0$$

$$\text{Product} = -3$$

$$\text{Difference} = 2$$

$$(+3, -1) \Rightarrow 3\sqrt{2}, -\sqrt{2}$$

Comparison of roots =

$$\begin{array}{l} \text{Roots of } x = 3\sqrt{2}, 5\sqrt{2} \\ \text{Roots of } y = 3\sqrt{2}, -\sqrt{2} \end{array} \left\{ \begin{array}{ll} 3\sqrt{2} = 3\sqrt{2} & 5\sqrt{2} > 3\sqrt{2} \\ 3\sqrt{2} > -\sqrt{2} & 5\sqrt{2} > -\sqrt{2} \end{array} \right\}$$

$$\boxed{\text{So, } x \geq y}$$

$$(I) x^2 - 5\sqrt{3}x + 18 = 0$$

$$x^2 - 5x + \frac{18}{3} = 0$$

$$x^2 - 5x + 6 = 0$$

$$(+3, +2)$$

$$(3\sqrt{3}, 2\sqrt{3})$$

$$(II) y^2 - 11\sqrt{2}y + 60 = 0$$

$$y^2 - 11y + \frac{60}{2} = 0$$

$$y^2 - 11y + 30 = 0$$

$$(+5, +6)$$

$$(5\sqrt{2}, 6\sqrt{2})$$

Comparison of roots →

Squaring all
Roots $\Rightarrow$

$$\begin{array}{l} x = 27, 12 \\ y = 50, 72 \end{array} \left\{ \begin{array}{ll} 27 < 50 & 12 < 50 \\ 27 < 72 & 12 < 72 \end{array} \right\}$$

$$\boxed{\text{So, } x < y}$$

$$(I) 7x^2 - 27\sqrt{5}x + 90 = 0$$

$$7x^2 - 27x + \frac{90}{5} = 0$$

$$7x^2 - 27x + 18 = 0$$

$$\text{Product} = 126$$

$$\text{Sum} = 27$$

$$(21\sqrt{5}, 6\sqrt{5}) \times 5$$

$$\{105\sqrt{5}, 30\sqrt{5}\}$$

$$(II) 5y^2 + 3\sqrt{7}y - 98 = 0$$

$$5y^2 + 3y - \frac{98}{7} = 0$$

$$5y^2 + 3y - 14 = 0$$

$$\text{Product} = 70$$

$$\text{Difference} = 3$$

$$(7\sqrt{7}, -10\sqrt{7}) \times 7$$

$$\{49\sqrt{7}, -70\sqrt{7}\}$$

Comparison :- Squaring all roots
 or Putting value of $\sqrt{5} = 2.24$
 $\sqrt{7} = 2.65$

$$x = 105(2.24), 30(2.24)$$

$$y = 49(2.65), -70(2.65)$$



$$\text{Roots of } x = 235, 67$$

$$\text{Roots of } y = 130, -185.5$$

$$\left\{ \begin{array}{ll} 235 > 130 & 67 < 130 \\ 235 > -185.5 & 67 > -185.5 \end{array} \right\}$$

There is both $x > y$
 and $x < y$

So, Relationship can not be established.

$$(I) x^2 + \sqrt{2}x - 4 = 0$$

$$x^2 + x - 2 = 0$$

$$\text{Product} = 2$$

$$\text{Difference} = 1$$

$$(-2, +1)$$

$$(-2\sqrt{2}, +\sqrt{2})$$

$$(II) y^2 - (5 + \sqrt{2})y + 5\sqrt{2} = 0$$

$$\text{Product} = 5\sqrt{2}$$

$$\text{Sum} = 5 + \sqrt{2}$$

$$(5, \sqrt{2})$$

Both positive

Comparison of Roots $\Rightarrow$

$$\text{Roots of } x = -2\sqrt{2}, \sqrt{2}$$

$$\text{Roots of } y = 5, \sqrt{2}$$

$$\left\{ \begin{array}{l} -2\sqrt{2} < 5 \\ -2\sqrt{2} < \sqrt{2} \end{array} \right.$$

$$\left\{ \begin{array}{l} \sqrt{2} < 5 \\ \sqrt{2} = \sqrt{2} \end{array} \right.$$

$$\sqrt{2} = \sqrt{2}$$

$$\boxed{\text{So, } x \leq y}$$

$$(I) x^2 + (2\sqrt{5} - 7)x - 7\sqrt{20} = 0$$

$$\text{or } x^2 - (7 - 2\sqrt{5})x - 7\sqrt{20} = 0$$

$$\text{Product} = 7\sqrt{20} = 14\sqrt{5}$$

$$\text{Difference} = 7 - 2\sqrt{5}$$

$$(-2\sqrt{5}, +7)$$

$$(II) y^2 + 6\sqrt{5}y + 40 = 0$$

$$\text{or } y^2 + 6y + 8 = 0$$

$$\text{Product} = 8$$

$$\text{Sum} = 6$$

$$(-4, -2) \Rightarrow (-4\sqrt{5}, -6\sqrt{5})$$

Comparison of Roots $\Rightarrow$

(Squaring all roots)

$$x = -2\sqrt{5}, +7$$

$$y = -4\sqrt{5}, -2\sqrt{5}$$

$\Downarrow$

$$x = -(20), +49$$

$$y = -(80), -(20)$$

$$\left\{ \begin{array}{ll} -20 > -80 & +49 > -80 \\ -20 = -20 & +49 > -20 \end{array} \right\} \quad \boxed{x \geq y}$$

$$(I) x^2 - 3\sqrt{5}x + 10 = 0$$

$$x^2 - 3x + 2 = 0$$

$$\text{Product} = 2$$

$$\text{Sum} = 3$$

$$(2, 1) \Rightarrow (2\sqrt{5}, \sqrt{5})$$

$$(II) y^2 - (6 + \sqrt{3})y + 6\sqrt{3} = 0$$

$$\text{Product} = 6\sqrt{3}$$

$$\text{Sum} = 6 + \sqrt{3}$$

$$(6, \sqrt{3})$$

Comparison of Roots $\rightarrow$

Squaring all roots $\rightarrow$

$$x = (20), (5)$$

$$y = (36), (3)$$

$$\left\{ \begin{array}{ll} 20 < 36 & 5 < 36 \\ 20 > 3 & 5 > 3 \end{array} \right\}$$

Here are both $x > y$
and $x < y$

So, Relationship between both x and y
can not be established.